

Ratio and fraction

Moving between “three parts to five” and “three eighths of the whole”.

LESSON 42–43

2 × 40 MIN

MATEMATIKA 6, §9

S7 12

LESSON CLOCK

20'

25'

25'

8'

2

From ratio to fraction	20 min
From fraction to ratio	25 min
One quantity from the other	25 min
Mixed problems	8 min
Homework	2 min

LEARNING OBJECTIVES

- Write each part of a ratio as a fraction of the whole.
- Write a ratio from two fractions of a whole.
- Find one quantity from the other using the ratio.
- Solve problems mixing ratios and fractions.

TEXTBOOK

National	pp. 113–120
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.2

01

Explanation

From ratio to fraction

Add the parts of the ratio; each part is that many of the total.

RATIO	TOTAL PARTS	FIRST AS A FRACTION	SECOND AS A FRACTION
2 : 3	5	$\frac{2}{5}$	$\frac{3}{5}$
3 : 5	8	$\frac{3}{8}$	$\frac{5}{8}$
1 : 4	5	$\frac{1}{5}$	$\frac{4}{5}$
2 : 3 : 5	10	$\frac{2}{10} = \frac{1}{5}$	$\frac{3}{10}$ and $\frac{5}{10}$

THE DENOMINATOR IS THE TOTAL, NOT THE OTHER PART

In 2 : 3 the first quantity is $\frac{2}{5}$ of the whole — but $\frac{2}{3}$ of the second quantity. Both statements are true and they mean different things.

From fraction to ratio

If one part is $\frac{3}{8}$ of the whole, the rest is $\frac{5}{8}$, so the ratio is 3 : 5.

GIVEN	THE REST	RATIO
$\frac{3}{8}$ are boys	$\frac{5}{8}$	3 : 5
$\frac{2}{7}$ are red	$\frac{5}{7}$	2 : 5
$\frac{1}{3}$ is water	$\frac{2}{3}$	1 : 2
$\frac{3}{4}$ is full	$\frac{1}{4}$	3 : 1

A COMMON DENOMINATOR TURNS FRACTIONS INTO A RATIO

If $\frac{1}{2}$ is juice and $\frac{1}{3}$ is water, write both over 6: $\frac{3}{6}$ and $\frac{2}{6}$, so the ratio is 3 : 2 — with a sixth left over for something else.

One quantity from the other

PROBLEM	WORKING	ANSWER
boys to girls 2 : 3, 14 boys	one part = 7	21
ratio 3 : 5, the smaller is 12	one part = 4	20
ratio 4 : 7, the larger is 35	one part = 5	20
ratio 2 : 3, 14 boys: the class	$5 \cdot 7$	35

FIND ONE PART FIRST, EVERY TIME

Divide the known quantity by its own number in the ratio. That single number then answers every other question in the problem.

Mixed problems

PROBLEM	ANSWER
a drink is 1 part syrup to 4 water; syrup as a fraction	$\frac{1}{5}$
600 ml of that drink: the syrup in it	120 ml
$\frac{3}{8}$ of a class of 32 are boys: the ratio of boys to girls	3 : 5
and the number of girls	20

READ WHICH QUANTITY THE GIVEN NUMBER BELONGS TO

“Fourteen boys” is one part of the ratio; “a class of thirty-five” is the total. The two lead to different first steps, and misreading which is given accounts for most lost marks here.

02

Worked examples

EXAMPLE 1 Boys to girls is 2 : 3 and there are 14 boys. How many girls, and how many pupils?

1

14 boys is 2 parts, so one part is 7.

2

Girls: $3 \cdot 7 = 21$.

3

Class: $5 \cdot 7 = 35$.

Check: $14 + 21 = 35$ ✓

ANSWER

21 girls, 35 pupils

EXAMPLE 2 In a class $\frac{3}{8}$ are boys. Write the ratio of boys to girls.

① Girls are $1 - \frac{3}{8} = \frac{5}{8}$.

② Both are eighths, so compare the numerators.

③ $3 : 5$

ANSWER

$3 : 5$

EXAMPLE 3 A drink is 1 part syrup to 4 parts water. How much syrup is in 600 ml?

① Total parts: 5.

② Syrup is $\frac{1}{5}$ of the drink.

③ $600 \div 5 = 120$ ml.

ANSWER

120 ml

03 Interactive model

Mix a jug of squash in front of the class at 1 : 4 and ask for the fraction that is syrup; the fifth is visible in the jug.

Ratio to fraction and back

In 2 : 3 the first part is what fraction of the whole?

$$\frac{2}{3}$$

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{2}$$

In 3 : 5 the second part is:

$$\frac{5}{8}$$

$$\frac{5}{8}$$

$$\frac{3}{8}$$

$$\frac{5}{3}$$

If 3 8 are boys, the ratio of boys to girls is:

3 : 8

3 : 5

5 : 3

8 : 3

Boys to girls 2 : 3 with 14 boys gives girls:

7

21

28

35

And a class of:

21

28

35

42

A drink 1 : 4 syrup to water is what fraction syrup?

$$\frac{1}{4}$$

$$\frac{1}{5}$$

$$\frac{4}{5}$$

$$\frac{1}{3}$$

600 ml of it contains syrup:

100 ml

120 ml

150 ml

480 ml

In 2 : 3, the first is what fraction of the second?

$$\frac{2}{5}$$

$$\frac{2}{3}$$

$$\frac{3}{2}$$

$$\frac{3}{5}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Fraction of the whole	Butunning qismi	Часть от целого
Total parts	Jami ulushlar	Всего частей
One part	Bir ulush	Одна часть
To convert	O'girmoq	Преобразовать
Share	Ulush	Доля
Proportion	Proporsiya	Пропорция
Equivalent	Teng kuchli	Равносильный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To turn a ratio into fractions you:

multiply the parts

add the parts

subtract them

divide them

In $1 : 4$ the first part is:

$$\frac{1}{4}$$

$$\frac{1}{5}$$

$$\frac{4}{5}$$

$$\frac{1}{3}$$

If 2 7 are red, the ratio red to the rest is:

$2 : 7$

$2 : 5$

$5 : 2$

$7 : 2$

Ratio 3 : 5 with the smaller 12 gives the larger:

15

20

24

36

Ratio 4 : 7 with the larger 35 gives the smaller:

15

20

25

28

The first step in every such problem is:

add the parts

find one part

divide the total

multiply

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. In $2 : 3$ the first as a fraction

ANSWER $\frac{2}{5}$

2. In $3 : 5$ the second as a fraction

ANSWER $\frac{5}{8}$

3. In $1 : 4$ the first as a fraction

ANSWER $\frac{1}{5}$

4. If $\frac{3}{8}$ are boys, the ratio boys : girls

ANSWER $3 : 5$

5. If $\frac{2}{7}$ are red, the ratio red : rest

ANSWER $2 : 5$

6. The parts of $2 : 3 : 5$

ANSWER 10

7. In that ratio the last as a fraction

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. Boys to girls 2 : 3, 14 boys: girls

ANSWER 21

2. And the class

ANSWER 35

3. Ratio 3 : 5, smaller 12: larger

ANSWER 20

4. Ratio 4 : 7, larger 35: smaller

ANSWER 20

5. A drink 1 : 4: syrup in 600 ml

ANSWER 120 ml

6. $\frac{3}{8}$ of 32 are boys: the girls

ANSWER 20

7. And the ratio boys : girls

ANSWER 3 : 5

Hard · stretch and reason

7 PROBLEMS

1. If $\frac{1}{2}$ is juice and $\frac{1}{3}$ water, the ratio juice : water

ANSWER 3 : 2

2. Ratio 5 : 3, the difference is 16: the total

ANSWER 64

3. Ratio 2 : 5, the total is 63: the parts

ANSWER 18 and 45

4. In 2 : 3, the first is what fraction of the second?

ANSWER $\frac{2}{3}$

5. A mixture 2 : 3 : 5 of 400 g: the three parts

ANSWER 80, 120, 200 g

6. If boys are $\frac{4}{9}$ of a class, the ratio boys : girls

ANSWER 4 : 5

7. Why is the denominator the total and not the other part?

ANSWER A fraction of the whole counts every pupil

07 Homework

Homework — 5 tasks

Find one part first; every other number in the problem follows from it.

- In 3 : 7 write each part as a fraction of the whole.
- If $\frac{2}{5}$ of a class are boys, write the ratio of boys to girls.
- Boys to girls is 3 : 4 and there are 18 boys. Find the girls and the class size.

4. A drink is 2 parts syrup to 7 water. How much syrup is in 900 ml?
5. A ratio is 5 : 8 and the larger quantity is 40. Find the smaller and the total.